

PalmMamba: Palm Intrinsic Features Learning Selective State Space Model for Palmprint Image Denoising

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Abstract—Palmprint-based biometric recognition has gained widespread attention due to its rich features, contactless acquisition, and low invasiveness. However, most existing methods neglect image quality, making them less effective for low-quality, noisy palmprint images. In this paper, we propose a palm intrinsic features learning selective state space model (PalmMamba) for palmprint image denoising, which consists of shallow feature representation, noise-insensitive palmprint-specific feature learning, and sharp palmprint image restoration modules. First, we convert the degraded noisy palmprint image into a high-dimensional shallow feature representation through a single-layer convolution backbone. Then, we develop parallel learning branches, including a second-order attention-based selective state space model and a mixed difference convolution module, to exploit diverse palmprint-specific features with both global and local details. Finally, we map the fine-grained palmprint-intrinsic feature map into the identity-preserved sharp palmprint image via a commonly used convolution layer. Extensive experimental results on five public palmprint databases demonstrate the encouraging performance of the proposed PalmMamba in palmprint image denoising.

Index Terms—Palmprint feature learning, palmprint image denoising, selective state space model, second-order attention.

I. INTRODUCTION

PALMPRINT conveys diverse specific characteristics, such as principal lines, wrinkles, and ridge-based texture patterns, which are unique and stable for an individual over a long

Received 21 October 2024; revised 19 January 2025; accepted 7 February 2025. Date of publication 14 August 2025; date of current version 21 October 2025. This work was supported in part by the National Key R&D Program of China under Grant 2023YFF0905602, in part by the National Natural Science Foundation of China under Grant 62176066 and Grant 62106052, and in part by the Natural Science Foundation of Guangdong Province under Grant 2023A1515012717. The associate editor coordinating the review of this article and approving it for publication was Dr Jongyoo Kim. (Corresponding authors: Lunke Fei; Bob Zhang.)

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Digital Object Identifier 10.1109/TMM.2025.3599093

period of time. Palmprint images can be easily captured by using normal cameras in a contactless manner, allowing for easy self-positioning, high hygiene, and low invasiveness. Consequently, palmprint recognition has received increasing research attention for biometric recognition in recent years, with several methods proposed in the literature [1], including hand-crafted coding descriptors [2] and learning-based methods [3]. In addition to academia, there is growing interest in palmprints within industry communities, such as Amazon’s “Amazon one” and Tencent’s “WePalm”, which use palmprints for personal authentication in electronic payment applications. Notably, most existing palmprint studies focus primarily on discriminative feature learning and matching, which are often based on the assumption that palmprint images are high quality and contain less noise. However, this assumption cannot be guaranteed in practical applications because palmprint images are typically collected in open or outdoor environments with complex background variations and illumination changes. As a result, unlike images captured under controlled laboratory conditions, the palmprint images collected in real-world settings often contain various types of noise, rendering most existing palmprint descriptors less effective.

Notably, with the aim of removing noisy parts from images to improve quality, several image denoising algorithms have been proposed in the literature. The typical methods include spatial-based, transform-based, and spatial-transform fusion methods [4]. Spatial-based denoising methods generally use the clean pixel information of an image as a reference to restore noisy patches. For example, Buades et al. [5] proposed a nonlocal means image denoising method that restores noisy regions with similar counterparts of the denoised pixels. In contrast, transform-based denoising methods typically convert the image from the spatial domain into the frequency domain and then filter out the noise-specific frequencies for image denoising [6]. For example, Mallat et al. [7] decomposed noisy images into different scales and frequencies via a wavelet transform, removing noise-oriented high-frequency information for image denoising. To leverage information from both the spatial and transform domains, Dabov et al. [8] proposed a block-matching and 3D filtering fusion method, which first groups image patches with similar structures and then performs 3D transformation and thresholding for image denoising. While these methods have

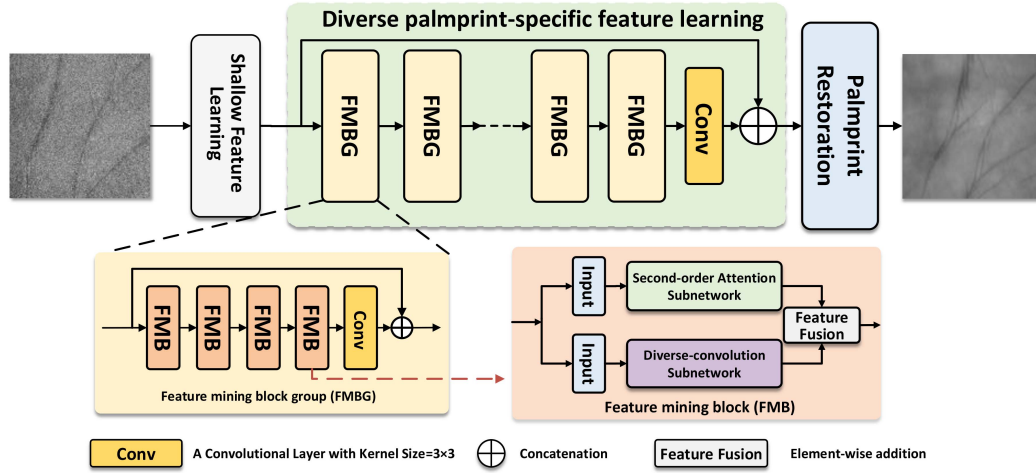


Fig. 1. An overview of the proposed PalmMamba, comprising a shallow feature representation backbone, a diverse palmprint-specific feature learning subnetwork, and a sharp palmprint image restoration module.

achieved reasonably good denoising results, they often require complex optimization algorithms and manual adjustment of parameters during the denoising process, leading to uncertainty in the final denoising performance.

Recently, with the encouraging breakthroughs of deep convolutional neural networks in image processing and computer vision, several deep learning-based methods have been proposed for image denoising [9], [10]. For example, Zhang et al. [11] proposed a CNN-based denoising method called the DnCNN, which employs multiple convolutional layers with residual connections and batch normalization to remove Gaussian noise from images. Zhang et al. [12] proposed a fast and flexible convolutional denoising network, FFDNet, which uses spatially invariant convolutional layers and downsampling for improved image denoising. Gou et al. [13] proposed a convolutional blind denoising network (CBDNet) that adaptively estimates noise levels for better noise removal. In addition, Yan et al. [14] proposed a black-box palmprint template reconstruction method based on the modified Progressive GAN (ProGAN) for reconstructing high-quality palmprint images. More recently, several studies [15], [16] have leveraged the superior global modeling capability of the transformer architecture to capture the long-distance dependencies in images for image denoising, yielding encouraging performance. For example, Liang et al. [15] proposed a powerful baseline, SwinIR, which leverages a shift window transformer with self-attention to extract image global dependencies to track denoising tasks, whereas Tian et al. [16] proposed CT-Net, which employs an interactive transformer to enhance the relationships between pixel features, thereby obtaining a more comprehensive global representation for improved image denoising performance. While these deep learning methods have achieved impressive image denoising results, they typically focus on learning the common features of images with widely used convolutional layers and the self-attention mechanism in transformers. Notably, palmprints intrinsically consist of ridge-based patterns with diverse wrinkles, making it difficult for most existing methods to capture the details necessary for high-quality palmprint image restoration.

In this paper, we propose a new palmprint feature learning selective state space model, named PalmMamba, for palmprint image denoising by specifically learning the intrinsic features of palmprints. Fig. 1 shows the basic framework of the proposed PalmMamba, which consists of a shallow feature learning backbone, diverse palmprint-specific feature learning, and sharp palmprint image restoration modules. First, we convert the noisy palmprint image into its high-dimensional shallow feature representation via a convolution layer. Then, we develop a stack of feature mining blocks (FMBs) to exploit diverse palmprint-specific features. In each FMB, we design a selective state space model (SSMs) combined with second-order channel attention (SCA) to capture the global long-path features, while in parallel, we form diverse difference convolution layers to repair short-path local texture features of the palmprint image. By doing so, we obtain a fine-grained palmprint-specific feature descriptor, which is further converted into a latent sharp palmprint image through a convolutional layer. Extensive experimental results on five widely used palmprint databases clearly demonstrate the effectiveness of the proposed method for palmprint image denoising.

The main contributions of our work are summarized as follows:

- We propose a new palmprint image denoising network that learns noise-insensitive, palmprint-specific features. To the best of our knowledge, this is the first work aimed specifically at studying palmprint image denoising for reliable palmprint recognition.
- We develop a second-order channel attention-embedded selective state space model to capture global palmprint features, while simultaneously designing diverse difference convolution layers to decouple noise information from palmprint texture patterns. This approach results in a comprehensive fine-grained palmprint feature representation for sharp palmprint image restoration.
- Our proposed PalmMamba can recover identity-preserved palmprint images with high visual quality from degraded,

noisy samples of the widely used PolyU, TJU, CASIA, BJTU, and XJTU-UP databases.

The remainder of this paper is organized as follows: Section II provides a brief review of the related work. Section III details the proposed PalmMamba for palmprint image denoising. Section IV analyzes the experimental results, and Section V concludes this paper.

II. RELATED WORK

In this section, we briefly review three related topics: palmprint image noise, the attention mechanism for image denoising, and the state space model.

A. Palmprint Image Noise

Image noise is the redundant disturbing information embedded in an image, which usually affects its visual quality and clarity. Typical image noise primarily includes Gaussian, Poisson, pepper, and real-world noise [17]. Gaussian image noise [18] is random noise following a normal distribution, which is typically caused by circuit noise within the camera or unstable environmental conditions during image capture. In contrast, Poisson image noise [19] follows a Poisson distribution and is commonly found in imaging scenes with low light conditions. Generally, Gaussian and Poisson noise are among the most widely used types of image noise in image denoising because of their well-defined statistical properties and ease of modeling when synthesizing corresponding noisy images [18]. Different from image noise that follows a specific distribution, pepper noise [20] is completely sparse and randomly distributed as white and black dots in an image, usually occurring during data transfer and affecting certain pixel values. In addition, the noise statistics of real-world palmprint images are usually complex and unknown due to various influencing factors such as imaging devices and lighting conditions. As a result, real-world image noise [13] is a mixed type of noise composing of various kinds of noises caused by different sources. To our knowledge, there have been two widely used approaches simulating the noise of real-world images. For example, Byun et al. [21] proposed an FBI-Denoiser network to model the signal-dependent and signal-independent real-world noisy images by combining the distributions of Gaussian and Poisson noise, and Guo et al. [13] proposed a realistic noise model to generate real-world noisy images via image signal processing. In this work, we simulate different levels of Gaussian, Poisson, and real-world noisy palmprint images for our palmprint image denoising study.

B. Attention Mechanism for Image Denoising

The attention mechanism enables the learning models to focus primarily on important features of an image while diverting less attention to irrelevant information through a dynamic weight adjustment strategy, providing benefits in various visual tasks, such as image classification and object segmentation [22], [23].

Motivated by these benefits, extensive studies have used attention mechanisms for image denoising tasks. Generally, the typical mechanisms include channel attention [24], spatial attention [25], hybrid attention [26], and self-attention [27]. Specifically, channel attention aims to automatically learn the weights of feature channels, enhancing important features while suppressing less important ones. For example, Zhao et al. [24] introduced a channel attention-based image denoising network that adaptively adjusts the feature channel representation through a residual learning strategy. Unlike channel attention, spatial attention focuses on important regions by generating a weight mask to enhance specific target areas of interest. For example, Wu et al. [25] proposed a spatial attention-based network that emphasizes learning the spatial relationships between convolutional features and filtering out less informative factors. By combining channel and spatial attention mechanisms, hybrid attention aims to enhance important features of images at both the channel and spatial levels. For example, Jang et al. [26] proposed a Spach transformer by fusing spatial-wise and channel-wise transformer blocks to restore image details from denoised blocks. In addition, self-attention computes the relevance (such as weights) of each element with respect to all other elements to capture global contextual information. For example, Wang et al. [27] proposed a Uformer image denoising network that uses nonoverlapping window self-attention to capture long-distance dependencies and adjust important image features. In this study, we develop a palmprint-specific selective state space model using parallel second-order channel attention and self-attention to extensively exploit the global feature relations of palms for palmprint image denoising.

C. State Space Model

The state space model (SSM) [28] excels at capturing long-range dependencies by mapping sequence data to a state space. As an improved version of the SSM, the Mamba deep learning model [29] has gained increasing attention because of its significant potential for long-range dependencies while maintaining linear computational complexity. In recent years, the SSM has been widely applied to various vision tasks, such as image classification [30], object detection [31] and image restoration [32]. Inspired by this wide use, many studies have applied SSM-based learning for image denoising. For example, Guo et al. [32] proposed MambaIR by employing a selective SSM to model long-range correlations for image restoration, whereas Fu et al. [33] proposed SSUMamba for hyperspectral image denoising by applying the bidirectional SSM to exploit long-range spatial-spectral dependencies. In addition, Liu et al. [34] developed forward and backward contiguous flattening SSM mechanisms, namely, HSIDMamba, to enhance spatial-spectral interactions for multispectral image denoising. In this study, we develop a selective SSM-based palmprint-specific feature learning network to exploit the long-range line-texture characteristics of palmprint images for a comprehensive contextual representation and for palmprint image denoising.

III. THE PROPOSED METHOD

In this section, we first present the overall pipeline of the proposed PalmMamba network and then elaborate on the palmprint-specific feature learning subnetworks for the recovery of latent sharp palmprint images.

A. Pipeline of the PalmMamba

In general, palmprints exhibit diverse specific characteristics, such as principal lines, wrinkles, and line-like texture patterns. Our proposed PalmMamba aims to extensively exploit the various palmprint-specific features from a degraded palmprint image and subsequently reconstruct a latent sharp palmprint image that retains intrinsic palmprint features.

To be more specific, given a degraded noisy palmprint image, I_N with the size of $H \times W \times C_{in}$, where H , W and C_{in} denotes the height, width, and channel numbers of the image, respectively, we first map it into the shallow feature representation via a 3×3 convolutional layer (denoted as $CONV_{3 \times 3}$) to efficiently describe a variety of visual features such as diverse textures and edges, as follows:

$$F_0 = CONV_{3 \times 3}(I_N), \quad (1)$$

where $F_0 \in R^{H \times W \times C}$ is the shallow feature representation of the palmprint image, and C represents the channel number of the shallow feature map. We then develop multiple groups of stacked feature mining blocks (FMBs), with each FMB consisting of parallel second-order attention and diverse-convolution subnetworks for exploring intrinsic palmprint features. To ensure that the translation of palmprint features is invariant, we add a 3×3 convolution layer at the end of each FMB group (FMBG) consisting of four cascaded FMBs. As a result, the palmprint-specific feature learning procedure of the FMBG can be expressed as follows:

$$F_{DF} = [CONV_{3 \times 3}(D_L^s(D_{L-1}^s(\dots D_1^s(F_0^s)))) + F_0^s]^S + F_0, \quad (2)$$

where D_l^s ($l = 1, 2, \dots, L, s = 1, 2, \dots, S$) represents the operation of the l -th FMB in the s -th group. S indicates the total number of FMBGs, and L represents the total number of FMBs in each FMBG, which are empirically set to 4 in this study. F_0^s denotes the input feature map of the s -th FMBG. $F_{DF} \in R^{H \times W \times C}$ denotes the final learned palmprint-specific feature map. Finally, we recover the latent sharp palmprint image (referred to as $I \in R^{H \times W \times 3}$) from the palmprint-specific features map via a commonly used 3×3 convolutional layer, as follows:

$$I = CONV_{3 \times 3}(F_{DF}). \quad (3)$$

In the following section, we detail the second-order attention and diverse convolution subnetworks within each FMB for learning diverse palmprint-specific features at both the global and local levels.

B. Learning Global Palmprint Features

In general, a palmprint contains rich relatively long line-like dominant characteristics, such as principal lines and wrinkles,

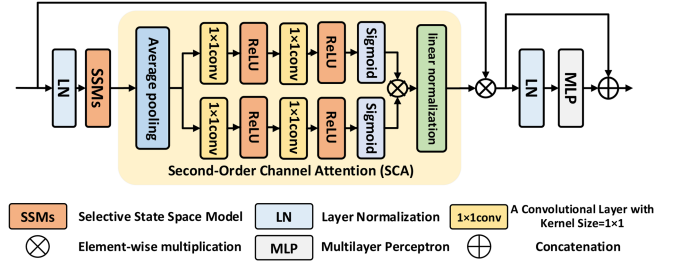


Fig. 2. The basic architecture of the second-order attention subnetwork designed for learning global palmprint features.

which convey important global structural information and discriminable palmprint-specific image features for palmprint image recovery. To capture this, we develop a second-order attention subnetwork, as shown in Fig. 2, to specifically exploit global features, including the spatial positional relationships and channel dependency of the global information in a palmprint image.

To be more specific, we first normalize the input feature map (denoted as F_{in}) using a layer normalization layer (referred to as LN) and use a selective state space model (referred to as $SSMs$) [29] to capture the long-range location-dependent information (denoted as F_1^{global}) of the palmprint image with linear computational complexity, as follows:

$$F_1^{global} = SSMs(LN(F_{in})). \quad (4)$$

In this way, we aim to capture global spatial information, including the relationships among distant pixels at related palmprint lines, and to establish a comprehensive structural understanding of the palmprint image.

Moreover, we design a second-order channel attention (SCA) mechanism, as shown in the middle part of Fig. 2, to enhance the global long-range dependency information in the channel dimension by dynamically adjusting the channel weights in the global feature map. Specifically, we use average pooling to aggregate the spatial information of the dependency feature map (i.e., F_1^{global}). Subsequently, we employ two parallel branches, each consisting of cascaded 1×1 convolution, rectified linear unit (ReLU), and sigmoid layers, to generate two groups of channel feature weights which we then fuse via element-wise multiplication and [0-1] linear normalization. As a result, we can formulate the adaptive channel weights (referred to as W) learning procedures as follows:

$$W = H_{norm}([H_{ada}(AvgPool(F_1^{global}))]^2), \quad (5)$$

where $AvgPool(\cdot)$ is the average pooling operation. $H_{ada}(\cdot)$ represents the parallel channel weight calculation branches, and $H_{norm}(\cdot)$ is the [0-1] linear normalization operation. By adaptively weighting the channel information, we can obtain an enhanced global palmprint feature map (denoted as F_2^{global}) that incorporates spatial positional relationships and channel dependency, as follows:

$$F_2^{global} = W * F_{in}. \quad (6)$$

Finally, we employ a normalization layer and a multilayer perceptron (MLP) with a residual connection to generate the

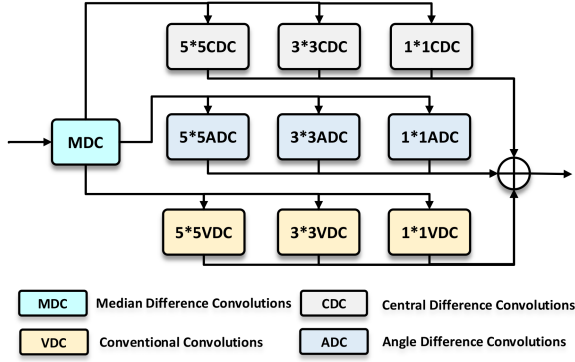


Fig. 3. The primary architecture of the diverse-convolution subnetwork designed for learning fine-grained palmprint texture features.

global palmprint-specific feature representation, referred to as $F_{global} \in R^{H \times W \times C}$, as follows:

$$F_{global} = MLP(LN(F_2^{global})) + F_2^{global}. \quad (7)$$

C. Texture Feature Refinement

In general, a palmprint image consists of various texture features, such as palm lines, wrinkles, and ridge patterns, which constitute the main characteristics of the palmprint. Learning different texture features of the palmprint is crucial for the visual quality restoration of a palmprint image. Existing studies [35] have demonstrated that center difference convolution and angle difference convolution can capture texture information from images, while most existing center difference convolution and angle difference convolution fail to capture the fine-grained texture features and the hierarchical line features unique to palmprint images due to the fixed kernels in the difference convolution layers. To address this, we develop a diverse-convolution subnetwork (DCS) with different scales of difference convolution, as shown in Fig. 3, to specifically exploit the local fine-grained texture features from a degraded palmprint image.

Notably, image noise often shares similar high-frequency information with palmprint texture patterns, making it difficult for most existing denoising methods to distinguish noise completely from irregular tiny texture patterns. In this study, we first employ a 3×3 median difference convolution [36] layer (denoted as $H_{MDC}(\cdot)$), which is commonly used to eliminate simple noise from an image, to remove the noise information from the input palmprint feature map (referred to as $F_{in}^{DCS} \in R^{H \times W \times C}$), as follows:

$$F_1^{DCS} = H_{MDC}(F_{in}^{DCS}), \quad (8)$$

where F_1^{DCS} is the denoised palmprint feature map. Moreover, we utilize a 5×5 central difference convolution and a 5×5 angle difference convolution to perceive the local texture features of the palmprint image, and we employ 1×1 and 3×3 convolutions to recover the texture details. Finally, we utilize the traditional convolutions with the sizes of 1×1 , 3×3 , and 5×5 to complement the translation invariance of the texture features at different scales, thereby improving their robustness.

To achieve this objective, we can reformulate the fine-grained texture feature (denoted as $F_2^{DCS} \in R^{H \times W \times C}$) learning process as follows:

$$\begin{aligned} F_2^{DCS} = & \sum_{K \in \{1,3,5\}} H_{CDC}^{K \times K}(F_1^{DCS}) \\ & + \sum_{K \in \{1,3,5\}} H_{ADC}^{K \times K}(F_1^{DCS}) \\ & + \sum_{K \in \{1,3,5\}} H_{VDC}^{K \times K}(F_1^{DCS}), \end{aligned} \quad (9)$$

where $H_{CDC}^{K \times K}$, $H_{ADC}^{K \times K}$ and $H_{VDC}^{K \times K}$ are the central difference convolution, the angle difference convolution and the conventional convolution with a size of $K \times K$ ($K = 1, 3, 5$), respectively.

IV. EXPERIMENTS

In this section, we simulate a variety of noisy palmprint images on five widely used palmprint datasets, including the PolyU, TJU, CASIA, BJTU, and XJTU-UP databases, to evaluate our proposed PalmMamba for palmprint image denoising. For our proposed PalmMamba, we empirically set the numbers of FMBs and FMBGs to 4, and adopt the L1 loss function for the network to calculate the pixel-wise difference between the denoised image and the noise-free image. To stabilize the training and accelerate convergence, we adopt the Adam optimizer with the hyperparameters $\beta_1 = 0.9$ and $\beta_2 = 0.9999$ to learn the optimal parameters, and set the initial learning rate to 2×10^{-4} with a reduced rate of 2 for every 50 epochs. In addition, we set the training times to 150 epochs, and use a Nvidia 3090 GeForce GTX GPU to improve the training efficiency. All experiments are implemented using PyTorch 2.1.1 and Python 3.10 run on Ubuntu 20.3.04.

A. Databases

The **PolyU** [37] multispectral palmprint image database contains 24,000 palmprint images collected from 250 volunteers using a multispectral palmprint image acquisition device under four different illuminations: blue, green, red, and near-infrared (NIR) lights. Each palm of a volunteer provides 12 palmprint images for each illumination spectrum. Consequently, the PolyU database contains four spectral subdatasets, referred to as PolyU-Blue, PolyU-Green, PolyU-Red, and PolyU-NIR, each consisting of 6,000 images from 500 different palms.

The **TJU** [38] palmprint image dataset consists of 12,000 images collected from 300 subjects using a specially designed contactless palmprint acquisition device in a pure black environment, with each palm of a subject contributing 20 palmprint images.

The **CASIA** [39] palmprint image dataset includes 5,502 palmprint images collected from 312 different individuals using a normal CMOS camera in an open environment with consistent illumination conditions, with 6 to 17 images acquired for each palm.

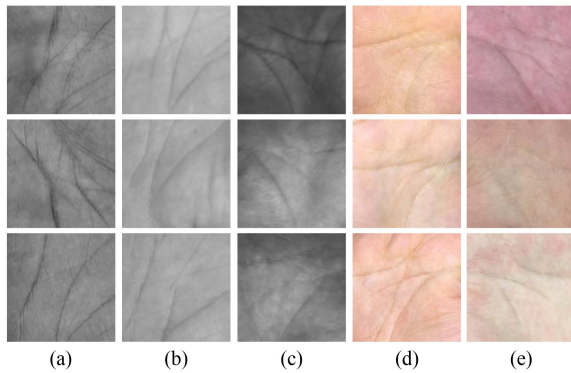


Fig. 4. The typical palmprint ROI images are randomly selected from the (a) PolyU, (b) TJU, (c) CASIA, (d) XJTU, and (e) XJTU-UP databases.

The **BJTU** [40] palmprint image dataset contains 2,663 palmprint images collected from 148 different volunteers in both indoor and outdoor scenarios using five different smartphones: an iPhone 6, Nexus 6p, Huawei Mate8, Nubia Z9, and Xiaomi Redmi 1S. Each volunteer captured approximately 3 to 5 palmprint images of each palm in the indoor and outdoor scenarios, respectively.

The **XJTU-UP** [41] palmprint image database contains 20,000 unconstrained palmprint images collected under both natural light and flash environments using five different acquisition devices: an iPhone 6S, LG G4, Huawei Mate8, Samsung Galaxy Note 5, and MI 8. Consequently, the XJTU-UP dataset comprises 10 subdatasets, each containing 2,000 palmprint images from the hands of 100 volunteers, with 10 images from each hand. In this study, we use the iPhone 6S and Huawei Mate8 subdatasets taken under natural light, referred to as XJTU-UP-iPhone and XJTU-UP-Huawei, respectively, for our palmprint image denoising experiments.

In this experiment, all palmprint images are cropped into ROIs [38] with the sizes of 64×64 pixels. Fig. 4 shows some typical palmprint ROI images randomly selected from different datasets.

B. Results of Denoising Gaussian Noisy Palmprint Images

To evaluate the proposed method for denoising images with widely observed Gaussian noise, we add three levels of Gaussian noise with standard deviations of 15, 25, and 50 (i.e., $\sigma = 15, 25, 50$) to the palmprint images from the PolyU-Green, PolyU-Red, TJU, CASIA, BJTU, XJTU-UP-Huawei, and XJTU-UP-iPhone datasets, respectively. For grayscale palmprint image denoising, the PolyU-Blue and PolyU-NIR palmprint samples are used for training, whereas the PolyU-Green, PolyU-Red, TJU, and CASIA samples are used for denoising testing. For RGB-based palmprint image denoising, such as BJTU, XJTU-UP-iPhone, and XJTU-UP-Huawei, we randomly select 20 palmprint images for each palm from each dataset as test samples and use the remaining images as training samples. For better evaluation, we compare the proposed method with seven state-of-the-art denoising methods, including [8], DnCNN [11], FFDnet [12], DCANet [42], SwinIR [15],

CTNet [16], and MambaIR [31]. We employ the widely used PSNR [43] and SSIM [44] as the denoising evaluation metrics. The PSNR evaluates the similarity of images by calculating the peak signal-to-noise ratio between them; the higher the value, the closer the denoising result is to the clean image. In contrast, the SSIM computes the similarity between images by evaluating the similarity of their structure and brightness, providing a way to assess image quality that aligns more closely with human perception.

Table I presents the denoising PSNR and SSIM results obtained by different methods, with the top two results marked in bold and underlined, respectively. The proposed PalmMamba method consistently outperforms traditional denoising methods (such as BM3D) and typical deep learning networks (such as the DnCNN and FFDNet) by achieving significantly higher PSNR and SSIM values. This is because our proposed method can exploit both long-distance global and fine-grained local palmprint-specific features to recover high-quality palmprint images, whereas most existing denoising methods rely heavily on the local self-similarity of the images and struggle to recover the tiny texture features of palmprint images. Our proposed method also achieves higher PSNR and SSIM values than channel attention-based denoising networks, such as DCANet and CTNet; the likely reason is that, unlike the one-order channel attention of existing methods, our proposed method employs second-order channel attention, which effectively enhances the interchannel dependency of palmprint global information, making the denoised palmprint images closer to the ground-truth samples. In addition, our proposed PalmMamba focuses on both location-dependent and channel-dependent information learning while capturing long-range dependencies resulting in better performance compared to other long-range dependency-based methods, such as SwinIR and MambaIR.

Fig. 5 shows denoised palmprint images restored using various methods. Our proposed PalmMamba demonstrates superior recovery of sharp palmprint images with clearer texture patterns compared to the other baselines, highlighting the promising image denoising performance of the proposed method.

C. Denoising Results of Poisson Noisy Palmprint Images

Poisson noise is commonly used to model photon noise in imaging sensors. To evaluate our proposed method for denoising palmprint images with Poisson noise, we add different levels of Poisson noise with peak values of 2, 4, 8, and 20 (i.e., $\kappa = 2, 4, 8, 20$) to the PloyU-Blue and PloyU-Green samples, simulating Poisson noisy palmprint images. The PloyU-Blue samples are used for training, whereas the PloyU-Green samples are used for testing. Table II presents the denoising PSNR and SSIM values computed by various methods, demonstrating that our proposed method achieves higher PSNR and SSIM values than five state-of-the-art methods, including the DnCNN [11], FFDNet [12], RIDNet [45], SwinIR [15], and HSDT [46]. Moreover, Fig. 6 presents typical denoised results of Poisson noisy palmprint images obtained via different methods, clearly showing that our proposed method produces sharper images that are

TABLE I
PSNR (dB) AND SSIM VALUES OBTAINED THROUGH VARIOUS METHODS FOR PALMPRINT IMAGE DENOISING ON THE POLYU-GREEN, POLYU-RED, TJU, CASIA, BJTU, XJTU-UP-HUAWEI, AND XJTU-UP-IPHONE DATASETS

Method	σ	PolyU-Green		PolyU-Red		TJU		CASIA		BJTU		XJTU-UP-Huawei		XJTU-UP-iPhone	
		RSNR	SSIM	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM
BM3D [8]	15	35.50	0.87598	37.63	0.90328	37.93	0.92643	38.32	0.93149	33.01	0.72305	36.64	0.87367	38.28	0.91032
DnCNN [11]		36.52	0.88218	39.16	0.92516	39.49	0.94299	39.84	0.94653	34.00	0.77762	37.21	0.88120	39.55	0.93822
FFDNet [12]		36.72	0.87869	39.69	0.92587	39.52	0.93970	39.94	0.94458	33.37	0.73585	35.66	0.85100	38.59	0.91873
DCANet [42]		36.96	0.88361	39.87	0.92788	40.03	0.94847	40.53	0.94763	33.79	0.74277	36.81	0.87608	38.70	0.92289
SwinIR [15]		36.99	0.88573	40.11	0.92977	40.47	0.94627	41.16	0.95217	34.13	0.79339	37.58	0.88933	39.94	0.94187
CTNet [16]		36.93	0.88056	40.00	0.92836	39.93	0.94769	40.48	0.94565	34.33	0.80389	37.54	0.88308	39.94	0.94274
MambaIR [31]		37.07	0.88725	40.19	0.92936	40.48	0.94639	41.18	0.95298	35.00	0.83689	37.80	0.89426	40.16	0.94403
Ours		37.11	0.89770	40.23	0.93118	40.51	0.94722	41.22	0.95445	35.06	0.84369	37.81	0.89524	40.24	0.94434
BM3D [8]	25	33.18	0.83028	35.42	0.89832	35.10	0.90484	35.78	0.91073	31.24	0.71246	34.68	0.85016	36.89	0.91188
DnCNN [11]		34.36	0.84339	36.73	0.91154	36.52	0.92236	36.73	0.92359	32.81	0.74449	35.87	0.85944	37.79	0.92277
FFDNet [12]		34.87	0.83930	38.45	0.91137	36.75	0.91746	36.94	0.92481	32.48	0.73276	35.66	0.85431	37.57	0.91974
DCANet [42]		35.10	0.84409	38.32	0.91213	38.40	0.92416	37.62	0.92776	32.74	0.73994	35.91	0.86074	37.86	0.92319
SwinIR [15]		35.42	0.84733	38.84	0.91553	38.63	0.92945	39.32	0.93789	33.03	0.74901	36.18	0.86320	38.25	0.92686
CTNet [16]		35.17	0.84519	38.76	0.91347	37.67	0.91972	37.54	0.92706	32.96	0.74541	36.10	0.86221	38.32	0.92706
MambaIR [31]		35.48	0.85741	39.16	0.91612	38.64	0.92986	39.34	0.93983	33.23	0.78008	36.30	0.86572	38.50	0.92928
Ours		35.60	0.86065	39.38	0.91624	38.66	0.93046	39.40	0.94105	33.54	0.78636	36.37	0.86863	38.53	0.92972
BM3D [8]	50	31.98	0.70727	35.63	0.80811	34.40	0.82137	33.81	0.81434	29.89	0.63463	32.83	0.85016	34.57	0.88843
DnCNN [11]		33.21	0.79164	36.31	0.88942	35.34	0.88859	35.94	0.90848	30.79	0.69453	33.27	0.82054	35.09	0.89235
FFDNet [12]		33.27	0.79167	36.62	0.89787	34.79	0.89667	34.92	0.91235	30.69	0.68583	33.65	0.82686	35.90	0.90548
DCANet [42]		33.32	0.79362	36.63	0.89342	35.32	0.90031	35.62	0.91356	30.77	0.69205	33.41	0.82056	35.47	0.89869
SwinIR [15]		33.57	0.80224	36.78	0.89635	36.09	0.90489	36.91	0.91935	31.69	0.71413	34.35	0.83429	36.14	0.90796
CTNet [16]		33.52	0.79852	36.72	0.89614	35.79	0.90224	35.68	0.91521	31.71	0.71638	34.33	0.83413	36.25	0.90914
MambaIR [31]		33.59	0.80273	36.79	0.89803	36.14	0.90539	36.94	0.91793	31.83	0.71849	34.58	0.83968	36.37	0.90986
Ours		33.64	0.80581	36.95	0.89974	36.22	0.90777	37.05	0.91991	32.00	0.72863	34.62	0.84038	36.37	0.91079

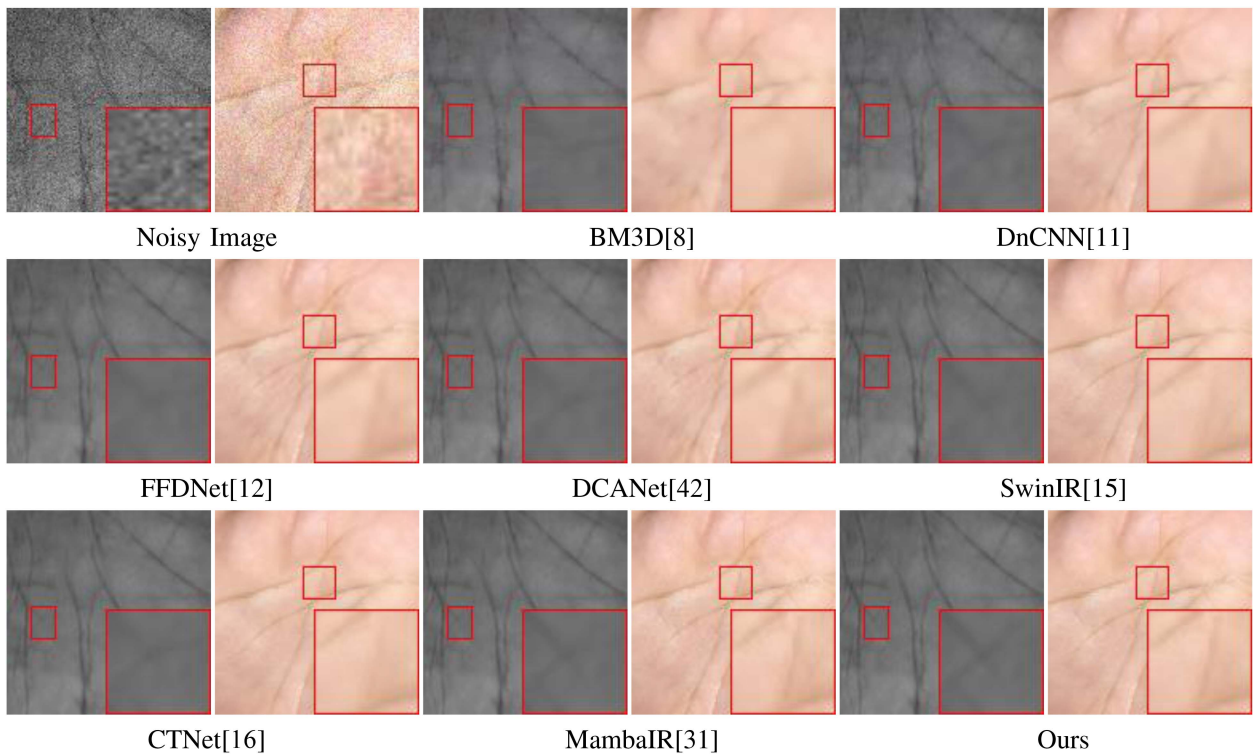


Fig. 5. Denoised results obtained by different methods for the PolyU-Red (left) and BJTU (right) palmprint images.

TABLE II
THE PSNR AND SSIM VALUES ARE OBTAINED VIA VARIOUS METHODS FOR DENOISING POLYU-GREEN PALMPRINT IMAGES AFFECTED BY POISSON NOISE

Method	$\kappa = 2$		$\kappa = 4$		$\kappa = 8$		$\kappa = 20$	
	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM
DnCNN[11]	37.32	0.89313	38.64	0.91898	40.14	0.94203	42.55	0.96644
FFDNet[12]	37.41	0.89506	38.71	0.91952	40.29	0.94309	42.63	0.96720
RIDNet[45]	37.46	0.89519	38.76	0.92049	40.32	0.94315	42.76	0.96704
SwinIR[15]	37.48	0.89614	38.77	0.92101	40.36	0.94375	42.74	0.96767
HSDT[46]	37.48	0.89579	38.78	0.92086	40.33	0.94355	42.80	0.96747
Ours	37.53	0.89665	38.82	0.92097	40.39	0.94419	42.83	0.96775

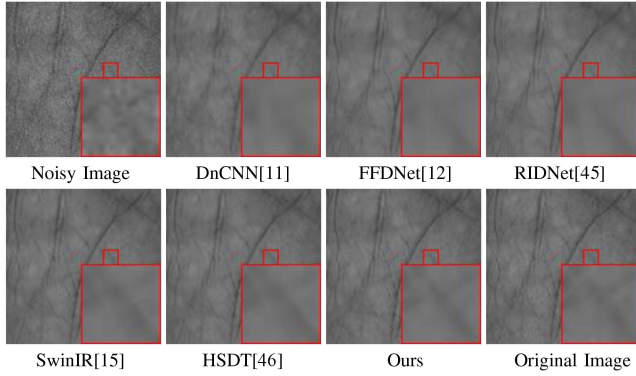


Fig. 6. Denoising results of Poisson noisy palmprint images obtained via different methods.

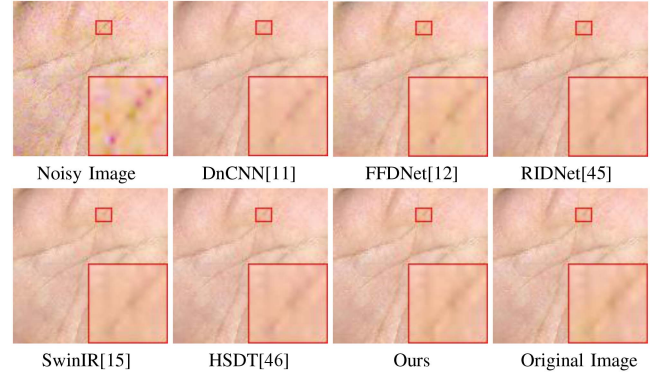


Fig. 8. Denoised results of a BJTU palmprint image affected by real-world noise, processed using different methods.



Fig. 7. (a) Two noise-free palmprint images, and (b) the corresponding noisy samples generated by realistic noise model.

more similar to the ground truth image; this indicates the promising effectiveness of the proposed method for denoising Poisson noisy palmprint images. This effectiveness is primarily due to the stronger spatial linkage capturing capability of our developed second-order channel attention and selective SSM, which better This decouples various types of noise for sharp image recovery.

D. Real-World Noisy Palmprint Image Denoising Results

To further evaluate our proposed method for denoising real-world noisy palmprint images, we add the simulated real-world noise generated by the widely used realistic noise model [13] to the BJTU, XJTU-UP-iPhone, and XJTU-UP-Huawei datasets. Fig. 7 shows some synthetically generated real-world noisy palmprint images by the realistic noise model. We then randomly select 70% of the palmprint images in each dataset as training samples and use the remaining images as test samples. Table III lists the PSNR and SSIM values of the denoising results obtained via the proposed method in comparison with

those of state-of-the-art networks, such as the DnCNN [11], FFDNet [12], RIDNet [45], SwinIR [15] and HSDT [46]. Our proposed method improves the PSNR and SSIM by approximately 0.67 dB and 0.00332, respectively, over the best values of the five comparison methods, specifically, HSDT. Moreover, Fig. 8 shows some representative denoised visualization results generated by different methods, demonstrating that our proposed method restores clearer texture patterns than the other methods. This improvement is due to our approach employing a diverse-convolution subnetwork to better exploit and preserve the palmprint-specific features during image denoising.

E. Ablation Studies

To evaluate the effectiveness of second-order channel attention (SCA) and the diverse-convolution subnetwork (DCS) for global and local feature learning, we removed each component from our PalmMamba framework to conduct comparative palmprint image denoising, as reported in Table IV. The proposed PalmMamba outperforms both variants when either the SCA mechanism or the DCS is removed. This is because the SCA mechanism compensates for the loss of spatial information by aggregating more global information, whereas the DCS exploits fine texture features to enhance local details. Consequently, fine-grained descriptions of both global and local palmprint characteristics can be restored using the proposed method, including both the SCA mechanism and the DCS, leading to improved PSNR and SSIM values.

TABLE III
PSNR AND SSIM VALUES OF DENOISING ON THE REAL-WORLD NOISY PALMPRINT IMAGES OBTAINED BY DIFFERENT METHODS ON THE BJTU, XJTU-UP-HUAWEI AND XJTU-UP-IPHONE DATASETS

Method	BJTU		XJTU-UP-Huawei		XJTU-UP-iPhone	
	RSNR	SSIM	RSNR	SSIM	RSNR	SSIM
DnCNN[11]	39.43	0.96725	43.97	0.98167	44.92	0.98114
FFDNet[12]	39.40	0.96703	44.01	0.98235	44.96	0.98264
RIDNet[45]	41.39	0.98011	44.76	0.98487	45.75	0.98459
SwinIR[15]	42.93	0.98486	45.82	0.98694	46.74	0.98641
HSDT[46]	42.80	0.98146	45.01	0.98522	46.08	0.98509
Ours	43.47	0.98684	46.67	0.98935	47.52	0.98841

TABLE IV
THE PSNR (DB) AND SSIM VALUES ARE OBTAINED FROM DIFFERENT PALMMAMBA VARIANTS ON THE POLYU-GREEN, POLYU-RED, TJU, AND CASIA DATASETS

Methods		σ	PolyU-Green		PolyU-Red		TJU		CASIA	
SCA	DCS		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
✓	×	15	37.06	0.88673	40.14	0.93029	40.49	0.94630	41.06	0.95107
×	✓		37.08	0.88695	40.13	0.93038	40.41	0.94569	41.02	0.95342
✓	✓		37.11	0.89770	40.23	0.93118	40.51	0.94722	41.22	0.95445
✓	×	25	35.56	0.85062	38.78	0.91540	38.63	0.92900	39.36	0.93868
×	✓		35.56	0.85075	38.81	0.91585	38.60	0.92709	39.39	0.93241
✓	✓		35.60	0.86065	39.38	0.91624	38.66	0.93046	39.40	0.94105
✓	×	50	33.61	0.80452	36.90	0.89803	36.05	0.90396	36.99	0.92016
×	✓		33.63	0.80578	36.90	0.89965	36.05	0.90346	37.02	0.92050
✓	✓		33.64	0.80581	36.95	0.89974	36.22	0.90777	37.05	0.92991

TABLE V
THE IDENTIFICATION ACCURACY ACHIEVED BY VARIOUS METHODS ON PALMPRINT IMAGES WITH DIFFERENT LEVELS OF GAUSSIAN NOISE AND SAMPLES RECOVERED BY OUR PROPOSED METHOD

Method	LLDP[49]	DRCC[50]	PCANet[51]	SDDLm[52]	ResNet[53]	ConvNeXt[54]
$\sigma = 15$	97.12%	97.62%	99.75%	95.13%	87.12%	83.00%
DeNoise	98.75%	97.75%	99.75%	95.75%	89.75%	87.62%
$\sigma = 25$	93.00%	97.37%	99.50%	95.12%	43.75%	68.50%
DeNoise	99.00%	97.75%	99.75%	96.00%	87.87%	87.37%
$\sigma = 50$	55.50%	95.62%	95.37%	91.87%	5.12%	46.87%
DeNoise	97.12%	97.12%	98.50%	95.25%	78.00%	79.37%

F. Palmprint Recognition Results

To evaluate the identity-preserved quality of denoised images obtained via our proposed method, we adopt six typical methods, including two representative palmprint descriptors, LLDP [47] and DRCC [48], and four feature learning networks: such as PACNet [49], SDDLm [50], ResNet [51], and ConvNeXt [52]. These methods are used to conduct palmprint recognition on the PloyU-Blue dataset. During palmprint recognition, we select the first two palmprint images without noise for each palm as the gallery set and use other images with varying levels of Gaussian noise, along with the corresponding denoised results, as the query set. In addition, for feature learning methods such as PACNet, SDDLm, ResNet, and ConvNeXt, we also use gallery samples to train the learning models. Table V presents the recognition results obtained by different methods on the noisy and denoised palmprint samples. All palmprint recognition methods achieve significantly higher identification rates on the denoised images than on the noisy images. This finding demonstrates that

our proposed denoising method can effectively preserve the discriminative features of palmprints during image denoising.

G. Parameter Analysis

Our proposed palmprint-specific feature learning module consists of multiple FMB groups, with each FMBG including several FMBs. Fig. 9 shows the PSNR and SSIM values of PalmMamba for different numbers of FMBs and FMBGs. It is observed that the PSNR and SSIM increase significantly as the number of FMBs and FMBGs increases from 1 to 4 and then increase slowly with fluctuations. This is because, as the number of FMBs and FMBGs increases, more palmprint-specific features can be extracted until no more beneficial features are found. Considering both the denoising performance and model size, we empirically set the numbers of both FMBGs and FMBs to 4 for our proposed PalmMamba network, making a better trade-off when both effectiveness and efficiency are concerned.

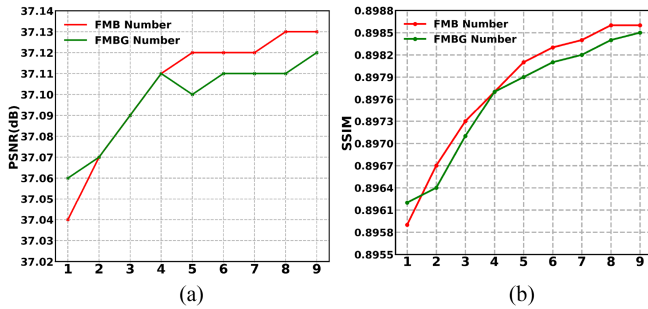


Fig. 9. The image denoising (a) PSNR and (b) SSIM values obtained by the proposed method with varying numbers of FMBs and FMBGs on the PolyU-Green noisy images, where $\sigma = 15$.

V. CONCLUSION

In this paper, we propose a simple yet effective PalmMamba network for palmprint image denoising. We first map the noisy palmprint image into a high-dimensional shallow representation. Next, we develop parallel feature learning branches, including the selective state space model combined with the second-order channel attention branch and the diverse difference convolution layer branch, to simultaneously exploit fine-grained palmprint-specific features at both the global and local levels. Finally, we restore the diverse palmprint intrinsic feature descriptors to latent sharp palmprint images via image convolution. Extensive experimental results on five benchmark databases clearly demonstrate the effectiveness of our proposed PalmMamba for denoising palmprint images with various types of noise. For future work, the extension of our proposed palmprint image denoising method can be explored for application to other palmprint-like biometric image denoising tasks, such as finger-knuckle-print (FKP) and palm-vein image denoising due to their similar hand-based characteristics as the palmprint.

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